

Transport Engineering Group
Department of Infrastructure Engineering
The University of Melbourne
Parkville, VIC 3010, Australia

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Editor-in-Chief
Transportation Research Part C: Emerging Technologies

Dear Editor-in-Chief,

Please consider the manuscript “GEL-Ped: Instability-aware guarded ensemble learning for short-horizon crowd prediction under flow shift” for publication in *Transportation Research Part C: Emerging Technologies*.

Short-horizon pedestrian forecasts support frequent replanning by mobile robots and automated vehicles, as well as local monitoring in crowded stations and terminals. The paper studies a narrow deployment problem: an interaction model trained on one flow can make a rare unstable correction when the encounter topology changes. GEL-Ped turns an observable property of the deployed predictor—local disagreement among independently initialized graph learners—into a conservative action. The graph ensemble remains fully responsible while its spread lies inside a leave-one-complete-run-out calibration envelope. Only an exceedance moves the forecast toward a bounded behavioural residual model.

The evaluation is organized by complete physical runs and deliberately separates a development-informed crossing stress test from cleaner external evidence. In perpendicular crossing, GEL-Ped improves all thirteen runs relative to the graph mean, with a 2.3% mean RMSE reduction and Holm-adjusted exact $p = 0.0037$. The median gain is smaller, and the manuscript diagnoses the influential unstable run rather than hiding it. A geometry-free coarse boosted model is marginally lower in crossing but 5–8% worse in the four other controlled regimes; GEL-Ped preserves graph accuracy there. On all five ETH/UCY leave-one-scene-out tests, the scene-macro change is negligible and non-significant. The paper therefore makes no universal accuracy claim.

The revision includes a five-member graph ensemble and robust median baseline; coarse, constant-velocity, learned-gate, Mahalanobis, and prospective-geometry controls; all fifteen predeclared Holm-adjusted tests; complete per-run values; collision-event counts; frozen threshold sensitivity; route-axis and future-label-filter audits; and recursive error analysis to 3.2 s. Code, immutable partitions, calibration records, processed complete-run outputs, tests, and figure scripts are available at github.com/AmirNetwork/GEL-Ped.

The manuscript is original, is not under consideration elsewhere, and all submission statements will be confirmed by the author. Thank you for your consideration.

Sincerely,

Amir Ghorbani
Transport Engineering Group
The University of Melbourne